Winkel Lösungen · mehr

1. $\boldsymbol{\varphi \approx 46{,}7°}$ $\cos\varphi = 4/\sqrt{34} \approx 0{,}686 \Rightarrow \varphi \approx 46{,}7°$

2. $\boldsymbol{\alpha \approx 38{,}7°}$ $\cos\alpha = |\begin{pmatrix}4\\0\\5\end{pmatrix} \cdot \begin{pmatrix}0\\0\\1\end{pmatrix}| \,/\, \sqrt{41} = 5/\sqrt{41} \Rightarrow \alpha \approx 38{,}7°$

3. $\boldsymbol{\varphi \approx 79{,}1°}$ $\overrightarrow{IL} = \begin{pmatrix}-4\\0\\4\end{pmatrix}, \overrightarrow{IJ} = \begin{pmatrix}-4\\6\\-2\end{pmatrix} \Rightarrow \cos\varphi = 8/(\sqrt{32} \cdot \sqrt{56}) \Rightarrow \varphi \approx 79{,}1°$